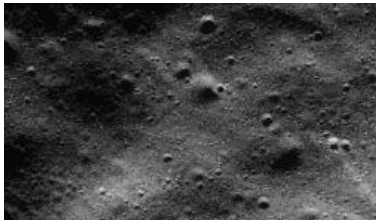
THE CHANDRAYAAN MOON MAPPING CHALLENGE

Team 21
Inter IIT Tech Meet 11.0

Problem Statement

1. To create AI/ML model, which may generate a very high spatial resolution image of the Moon from Low Resolution TMC Sensor
2. To create lunar atlas from generated HR Data.

TMC - 2 (5 m) → Deep Learning model → OHRC (30 cm)



Workflow

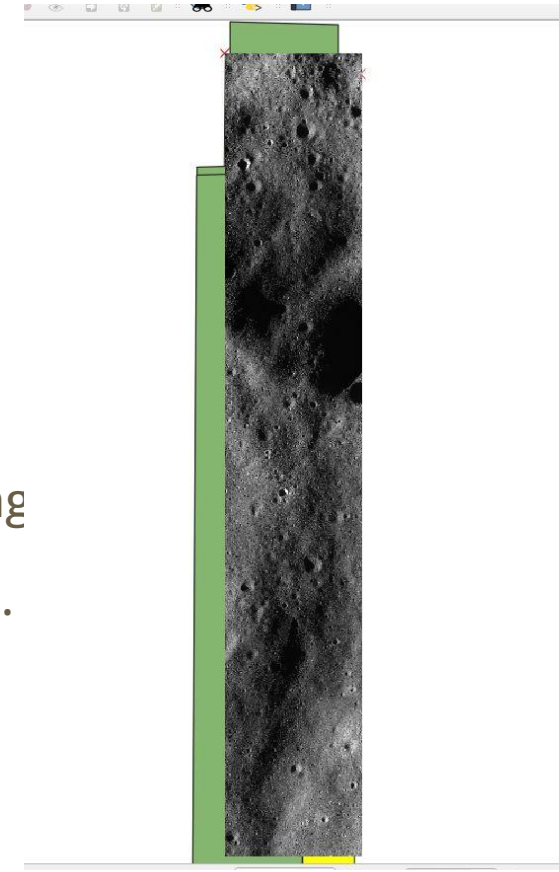
1. Downloaded TMC - 2 (5 m) and OHRC (30 cm) resolution from Website.

   https://pradan.issdc.gov.in/ch2/protected/payload.xhtml

2. Attached Lat Long to OHRC Img file using GDAL python
3. Finding the HR _ LR Overlap region using python script
4. Crop HR and LR data and divide it into Train , Validation and Test dataset.
5. Radiometric normalization of data before training.
6. Train GAN based super resolution model with our own loss function.
7. Mosaic generate HR data to create seamless atlas.

Data Preparation

1. Used QGIS for Shapefile Visualization.
2. Overlay Shapefile and Georeferenced TIFF Images.
3. TMC - 2 (5 m) LR - _nd* derived products with better geometric accuracy
4. OHRC (30 cm) HR - _nc* mens calibrated data in .Img format and converted it into georeferenced .Tif data.
5. Nearly 50 long Strips for overlapped data.



Data Preparation

6. Generate the overlapping area between LR and HR images using the shape files provided

7. Convert .Img images to georeferenced tiff images by adding the coordinates of each pixel for both LR and HR images

8. Crop the required overlapping area from both LR and HR images which makes our input and output for training the model

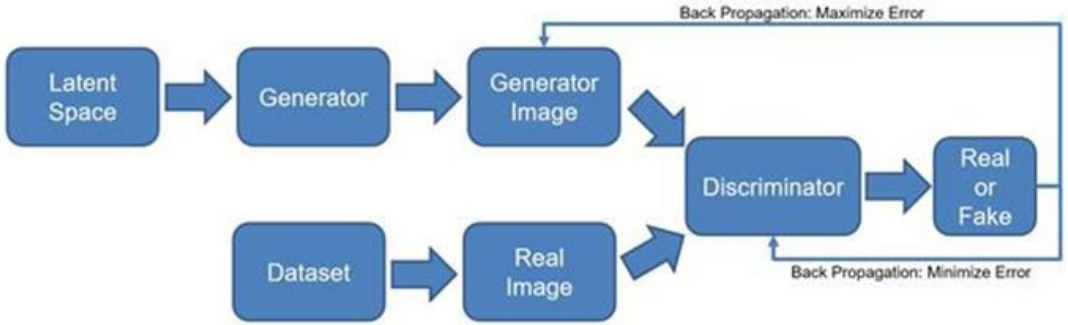
9. Model requires the use of specific dimension to be given as input and output for which we can create multiple tiles from same LR an HR image with respect to the dimension ratio between them

10. Data augmentation using rotation in 4 direction.

11. Radiometric Normalization of HR and LR Data before training.

Model - Modified SRGAN

1. The generator takes a low-resolution image and output a high-resolution image.
2. The discriminator check how real is the generated image.
3. In beginning, the generator produces bad quality high resolution images but as the training converges with multiple iterations, generator starts to produce realistic images.
4. At the end when training is completed, the discriminator can't differentiate between the original high-resolution image and the one created by the generator.

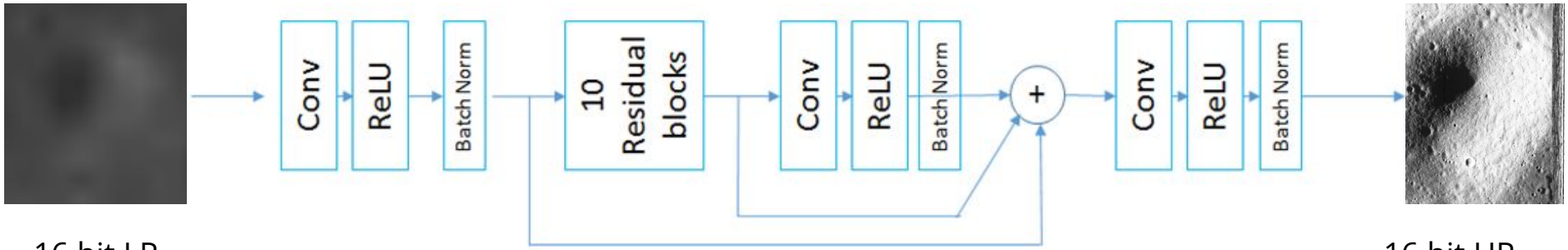


Model - Modified SRGAN

- We modified SRGAN model for satellite data.

The model consists of 3 main blocks:

1. Generator

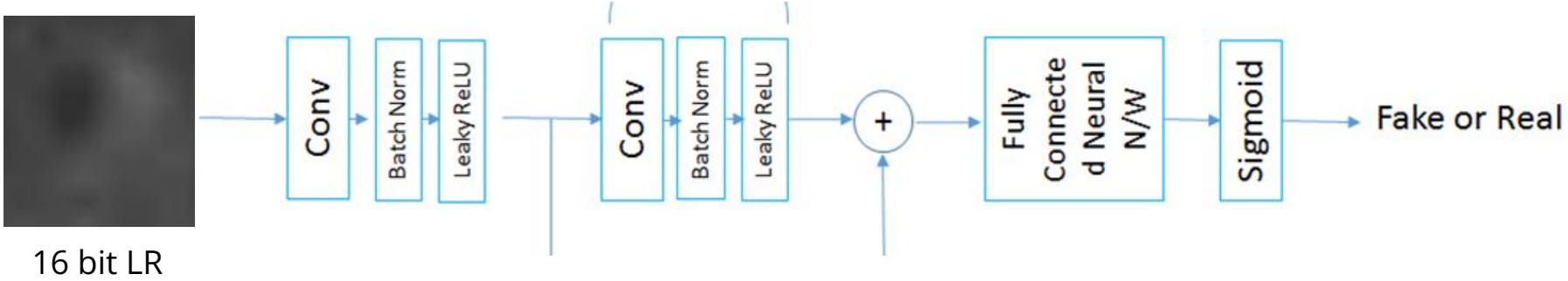


16 bit LR

16 bit HR

Model

2. Discriminator



3. Loss Function $\text{Total Loss function} = \text{MSE} + \lambda * \text{MGE}$

1. Combination of Mean Gradient Error (MGE) and Mean Square Error(MSE).
2. Formulated the loss function with the combination of Mean Gradient Error (MGE) and Mean Square Error(MSE).
3. MGE will provide loss w.r.t. sharpness of edges and MSE will provide the overall content loss.

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2$$

Challenges

(1) When LR and HR images are slightly miss registered, removal of max-pooling layer and pixel shuffler in last layers of model helps us to cater slight miss registration of few pixels in remote sensing imagery.

(2) Trained the model directly on 16-bit Moon data will increase the training time approximately by the factor of 3, as compare to train the model on 8 bit images.

(3) Moon data Projection - Selenographic coordinate system

Novelty

Modification w.r.t. basic SRGAN model, can be summarized as follows: -

(1) Layers is modified to take 16bit input image and gradient clipping is used in gradient explosion issue.

(2) ReLu was used in the last layer of generator to transform the classification problem into a regression problem.

(3) Max pool layers are removed as second last layer. It was observed that Maxpool layer removal has decreased the overall error and model converges faster.

(4) Pixel shuffler is removed from second last layer to handle miss registration of few pixels.

(5) The convolution kernel of the last layer of the generator was changed to 9×9.

(6) Loss function is changed to consider gradient change into account.

Hyper-parameters

Optimizer	ADAM	Gradient descent optimizer for training
Batch Size	32	Training batch size
α-Learning Rate for G and D	0.002	Same learning rate for G and D
Epoch	200	NO other stopping criteria during training
Drop out rate	10%	Training Drop out percentage
No of Datasets(256*256)	5000	Each of 256 * 256 (scan *pixels)
Hardware	Windows 10 with GP-GPU Tesla P100-PCIE Number of CUDA Cores- 3500,Onboard memory-16 GB,Computing Capability- 6.	
Framework	Python 3.7 , Pytorch and Tensorflow Cuda-11.3.	
Time to Train	4 hours	On P100 GPU

Quality Assessment

1. PSNR is to measure the similarity between two images

$$PSNR = 10 \times log(maxvalue^2/MSE)$$

2. The SSIM algorithm is used to test the similarity of two images, and its measurement or prediction of image quality
3. FSIM - FSIM value is between 0 and 1, where 1 is perfect feature similarity.

$$SSIM(X,Y) = \frac{(2u_X u_Y + C_1)(2\sigma_{XY} + C_2)}{\left(u_X^2 + u_Y^2 + C_1\right)\left(\delta_X^2 + \delta_Y^2 + C_2\right)} \quad (20)$$

where u_X and u_Y represent the means of the gray values of image X and image Y, respectively, and δ_X^2, δ_Y^2 and σ_{XY} represent the variances of the gray values of image X and image Y, respectively.

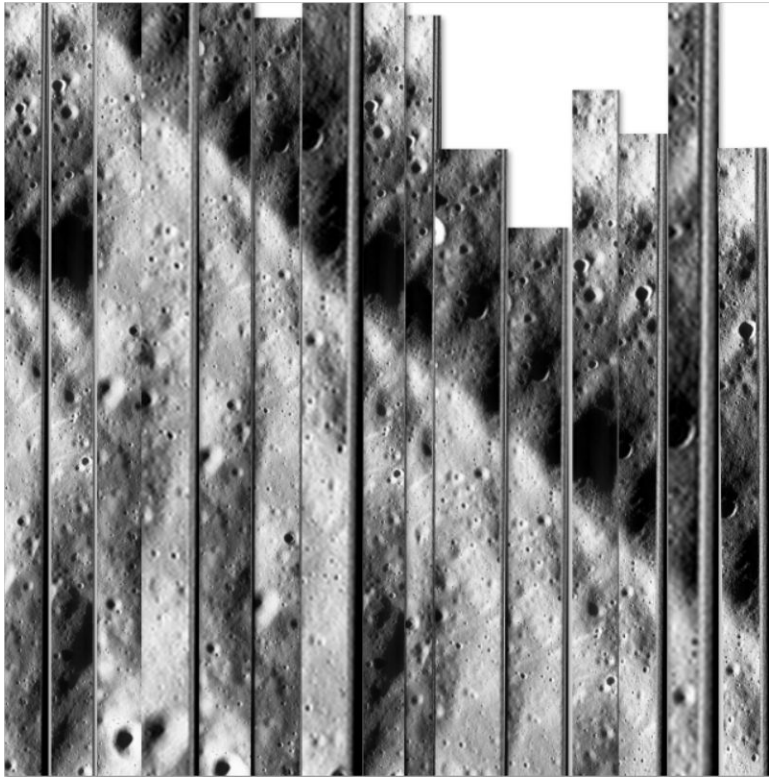
Generally, $C_1 = (K_1 * L)^2, C_2 = (K_2 * L)^2, K_1 = 0.01, K_2 = 0.03$ and L is the maximum image value.

Validation

Quality Matrices	PSNR	FSIM	SSIM
Ground Truth and Predicted data	18.16	0.78	0.75

Note - python Image Similarity Measures library was used for above calculations

Results - HD 30 cm Mosaic Map/ Atlas



Limitations :-

1. Model needs to be trained on more number of epochs. Due to resource limitation only 15 epochs was trained.
2. Model also introduced some artifacts.
3. Since the number of overlaps between Hr and LR was less , model can overfit easily .
4. Below results are the best area results.

Team

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